D 51670 (Pages : 3) Name.....

Reg. No.....

THIRD SEMESTER (CBCSS—UG) DEGREE EXAMINATION NOVEMBER 2023

B.Com./B.B.A

A11—BASIC NUMERICAL METHODS

(2019—2022 Admissions)

Time : Two Hours and a Half Maximum : 80 Marks

Answer should be written in English Only.

Part A

Answer **all** *questions.*

1. What is mean by an equation ?
2. Define a simultaneous equation in three variables.
3. Define a scalar matrix.
4. Define order of a matrix.
5. Show that $\begin{bmatrix} 2 & -1 & 3 \\ -1 & 2 & 1 \\ 3 & 1 & 4 \end{bmatrix}$ is symmetric.
6. Explain determinant of a 3 × 3 matrix with an example.
7. Define geometric progression and write the formula for finding n^{th} term of G.P
8. Define Harmonic progression.
9. Define immediate annuity.
10. What is mean by growing perpetuity ?
11. Define nominal rate of interest.
12. Explain the merits and demerits of mode.

Turn over

13. Define geometric mean.

14. Define mean deviation.

15. What are absolute measures of dispersion ?

(15 × 2 = 30, Maximum ceiling 25 marks)

Part B

Answer **all** *questions.*

16. Solve $14x - 28 + 2x - 4 = 6 + 2x - 10$.

17. Solve $x + y = 4$, $4x^2 - 3y^2 = 33$.

18. Demand for goods of an industry is given by the equation $pq = 100$ and supply is given by the equation $20 + 3p = q$ where p is the price and q is the quantity.

    Find p and q.

19. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, show that $A^2 - 4A - 5I = 0$.

20. Define arithmetic mean and also insert four arithmetic mean between 52 and 77.

21. If the 5th and the 10th terms of a G.P are 32 and 1024 respectively. Find the first term and the common ratio.

22. Find the compound interest Rs. 10,000 for 3 years at 5 % per annum.

23. Find the arithmetic mean of the following data :

Marks	:	10	20	30	40
No of students	:	40	32	12	5

(8 × 5 = 40, Maximum ceiling 35 marks)

Part C

Answer any **two** *questions.*

24. If $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 1 & 1 & 4 \end{bmatrix}$, show that $AA^{-1} = A^{-1} A = I$.

25. Solve the system of linear equation :

$x + y + z = 7$

$x + 2y + 3z = 16$

$x + 3y + 4z = 22$.

26. (a) Define annuity and explain different types of annuities.

(b) Find the total amount of annuity of Rs. 400 payable at the end of every quarter for 6 years at 8 % per annum compounded quarterly.

27. (a) Define quartile deviation and explain its merits and demerits.

(b) Using quartile deviation compare the following series and state which one is more variables ?

Series 1	:	5	10	27	90	38	56	29	43	39	86	30
Series 2	:	10	27	15	35	89	72	28	40	45	28	39

(2 × 10 = 20 marks)

D 51670-A **(Pages : 4)** **Name.....**

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A11—BASIC NUMERICAL METHODS

(2019—2022 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes	**Total No. of Questions : 20**	**Maximum : 20 Marks**

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

A11—BASIC NUMERICAL METHODS

(Multiple Choice Questions for SDE Candidates)

1. Given the term in the sequence 1, 3, 7, 15, 31, next term is :

   (A) 62. (B) 46.

   (C) 63. (D) 60.

2. One n^{th} term of a G.P. is ————————.

   (A) ar^n. (B) ar^{n-1}.

   (C) $a^n r$. (D) $a^{n-1} r$.

3. One common ratio of the G.P. $1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}$ ———————— .

   (A) 3. (B) $\frac{1}{3}$.

   (C) $\frac{1}{6}$. (D) 6.

4. If the sum of two numbers is 8 and their product is 15, numbers are :

   (A) – 5, – 3. (B) 5, 3.

   (C) 5, – 3. (D) 2, 6.

5. Find x if the number x, 7, 28 from a G.P. :

   (A) 4. (B) 0.

   (C) $\frac{7}{4}$. (D) $\frac{4}{7}$.

6. Find the sum of 1, 2, 3 ———————— 200.

   (A) 206. (B) 20100.

   (C) 2010. (D) 2,01,000.

7. Statistical result are :

   (A) Absolutely correct. (B) Not true.

   (C) True on average. (D) Universally true.

8. The primary data are collected by :

(A) Interview. (B) Observation.

(C) Questionnaire. (D) All these.

9. The classification 10–20, 20–30 is the example of :

(A) Exclusive. (B) Inclusive.

(C) Both. (D) None.

10. Histogram is a ————————.

(A) Diagram. (B) Graph.

(C) Pictogram. (D) Cartogram.

11. Bar Diagrams are :

(A) One dimensional. (B) Two.

(C) Three. (D) None of these.

12. ———————— filled by the informants.

(A) Questionnaire. (B) Schedule.

(C) Questionnaire or schedule. (D) All.

13. If the mid value of classes are 2.5, 7.5, 12.5, 17.5 the first class is :

(A) 2.5–7.5. (B) 2.5–2.5.

(C) 5–10. (D) 0–5.

14. When the upper limit of a class is not the lower limit of the next class, the series is known as :

(A) Individual. (B) Discrete.

(C) Inclusive (D) Exclusive.

15. Classification is the process of arranging data in ————————.

(A) Different columns. (B) Different rows.

(C) Different groups. (D) Columns and rows.

16. Frequency distribution is ————————.

(A) Variable. (B) Frequency.

(C) Class interval. (D) Table.

17. Lottery method is the example of :

(A) Systematic. (B) Quota.

(C) Unrestricted. (D) None.

18. ———————— refers to the column headings.

(A) Caption. (B) Stub.

(C) Body. (D) None.

19. ———————— are the designations of the row headings.

(A) Caption. (B) Stub.

(C) Caption or stub. (D) All of these.

20. ———————— is concerned with the amount of variation rather than with its direction.

(A) Skewness. (B) Kurtosis.

(C) Dispersion. (D) None.